



[Foreign Talent Management System] - Test Plan

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REVISION HISTORY

Revision History records the trail of revisions made to the document. Revision History is the responsibility of the Document Control Centre to ensure all the document revisions are justified and documented in accordance with proper document control procedures.

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REVISION HIGHLIGHTS

All changes recorded in this Revision Highlights will be for the latest revision only.

Clause No.	Description of Changes
All	Initial draft (v1.0) prepared based on FTMS High-Level Briefing dated April 2026. Process Flow and Timeline inputs pending; affected sections marked [PENDING].

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LIST OF ABBREVIATIONS

The following abbreviations shall have the following meaning unless otherwise stated

Abbreviation	Full Term
MDEC	Malaysia Digital Economy Corporation Sdn Bhd
BU	Business Unit
FTMS	Foreign Talent Management System
AMP	Amendment of approved application
AI	Artificial Intelligence
DP	Dependant Pass
EP	Employment Pass
FAT	Final Acceptance Test Testing
EXPATS	Expatriate Services (company-facing FTMS module)
FKW	Foreign Knowledge Worker
URS	User Requirement Specification
LTSVP	Long-Term Social Visit Pass
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
PII	Personally Identifiable Information
PVP	Professional Visit Pass
QA	Quality Assurance
RACI	Responsible, Accountable, Consulted, Informed
SIT	System Integration Test

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Abbreviation	Full Term
SRS	System Requirement Specification
SIT	System Integration Testing
SUT	System Under Test
TOE	Transfer of Endorsement (to new passport)
UAT	User Acceptance Testing
VCRM	Verification Cross Reference Matrix

DEFINITION OF TERMS

For the purpose of this document and unless the context obviously indicates another meaning, the following words and expressions shall have the following meaning:

Term	Definition
Cloud Computing	The delivery of computing services over the internet, including storage, databases, servers, networking, and software.
Service Level Agreement (SLA)	A contract between a service provider and a customer outlining the expected performance and responsibilities.
Pass	An official immigration permission allowing a foreign individual to enter, stay, work, or remain in Malaysia for a specific purpose and period.
Endorsement	The official act of recording an approved pass into the holder's passport, making the pass operational.
Foreign Knowledge Worker (FKW)	A foreign professional applying for or holding a work-related pass under the EXPATS module.
Principal Applicant	The primary FKW (typically EP holder) on whose pass dependants may apply for related passes.
Dependant	An eligible family member of a Principal Applicant who applies for a Dependant Pass or LTSVP.
Resubmission	A returned application sent back to the company/applicant for correction, requiring resubmission to re-enter the processing flow.
Decision Stage	The Meeting step in which an Approver makes the final Approve / Reject determination on an application.

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Term	Definition
Test Environment (TE)	A non-production environment that mirrors production configuration, used exclusively for executing test cases.
Defect	Any deviation between actual and expected behaviour identified during testing, classified by the Severity Level scale in 8.2.

REVIEW OF THIS DOCUMENT

This document will be reviewed every 1 year or when deemed necessary.

CONFLICTS BETWEEN DOCUMENTS

In the event of any conflict or inconsistency between this document and any other document, the document in the higher level of document hierarchy shall prevail.

COMPLIANCE

- a. All Heads of Division and Heads of Department are responsible to ensure that this Guideline is operationalized within their respective divisions and departments.

ASSOCIATED DOCUMENTS

This document should be read together with the following associated documents:

Title	Document ID
FTMS EXPATS Project Timeline / Schedule	FTMS-PT-EXPATS-001 [PENDING — to be received]
FTMS EXPATS System Requirement Specification	FTMS-SRS-EXPATS-001 [Reference]
FTMS EXPATS User Requirement Specification	FTMS-URS-EXPATS-001 [Reference]

1.0 INTRODUCTION

1.1. Purpose

This Test Plan defines the scope, approach, resources, schedule, and acceptance criteria for verification and validation of the **Foreign Talent Management System (FTMS) – EXPATS Module**.

The objective of this document is to provide a coherent and traceable basis for testing activities to enable and assist:

- the QA team in planning, designing, and executing test cases against the EXPATS module;
- the development team in identifying, fixing, and re-verifying defects raised during test execution;
- IT Operations and DevOps in preparing and supporting non-production environments;
- business stakeholders (companies, immigration officers, MDEC reviewers, MDEC management) in performing User Acceptance Testing (UAT);
- the maintenance team in supporting, sustaining, and enhancing the system once in production.

It is intended that this document forms the contractual basis for confirming that the EXPATS module satisfy its functional, non-functional, and compliance requirements prior to production release.

1.2. Scope

This Test Plan covers the EXPATS Module of FTMS, the company-facing one-stop service centre and digital application channel for managing Foreign Knowledge Worker (FKW) pass applications and related immigration processes in Malaysia. The following functional areas are covered:

- Company registration (Company Management)
- FKW application lifecycle for Principal and Dependant applicants (Application Management) across pass types: EP, DP, PVP, LTSVP
- Document upload, submission, and Return / Resubmission handling
- Payment 1 (pre-processing) and Payment 2 (post-approval)
- Internal Processing (Verification / Assessment)
- Meeting (Decision Stage) — Approval / Rejection
- Endorsement and Pass Issuance
- Subprocesses: Transfer of Endorsement (TOE) and Amendment (AMP)
- Supporting modules where invoked by EXPATS: Document Management, Master Data Management, Integration Layer, AI-Assisted Capabilities (Document Extraction, Document Validation, Decision Support)
- Functional, Integration, System, Performance, Security/Vulnerability, AI Component, and User Acceptance testing
- Migration verification — Document Migration (OSS → OSS) and Data Migration (MySQL → PostgreSQL), covered in detail under 5.1

The following are not included in this document:

1.	DE Rantau module (freelancer / digital nomad pass management)
2.	MTEP module (entrepreneur / startup founder pass management)
3.	Workflow-engine internals deep dive (covered separately under platform engineering)
4.	Technical architecture verification beyond integration contracts (covered separately under architecture review)
5.	Penetration testing engagements requiring specialist external vendors (subject to a separate scope of work)
6.	Test of legacy systems (System A / B / C) being decommissioned

1.3. References

The following documents and standards are referenced as inputs to this Test Plan:

No.	Document	Description
1.	FTMS Project Timeline	[PENDING – project schedule to be supplied. Will drive Section 3.0 Test Schedule milestones.]
2.	IEEE Std 829 — Standard for Software Test Documentation	Industry reference standard adopted for test plan structure and content.
3.	ISO/IEC/IEEE 29119 — Software Testing Standards	Industry reference standard for test process, documentation, and techniques.
4.	Workshop Slides	The presentation reference.

2.0 TEST ORGANIZATION AND RESPONSIBILITIES

The test organization for FTMS Phase 1 (FKW) is structured around a central QA function reporting to the Test Manager / QA Lead, with cross-functional support from Development, Business, Environment / DevOps, and Document Control. The roles and responsibilities for the test programme are as defined in S2.1 below, with accountability for each major test activity formalised in the RACI Matrix at S2.2.

2.1. Test Role and Responsibilities

The FTMS Phase 1 (FKW) test programme is delivered through a multi-organisational structure: MDEC (Client) provides programme governance, business expertise, and acceptance authority; TCS (Implementation Partner) provides delivery, build, and test execution capability; specialist external vendors are engaged for Penetration Testing. The roles and responsibilities are as follows:

2.1.1. MDEC Roles

Key Responsibilities

- Approves Test Plan; resolves escalated risks beyond Programme Manager authority; signs off Cutover Readiness Review and Go-Live Decision; owns business outcome of FTMS Phase 1.
- Owns business success of the programme; reviews phase closure reports; co-signs UAT acceptance; primary point of escalation for business-impact concerns.

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- Owns programme delivery from MDEC side; reviews Master Test Plan; resolves cross-team blockers; chairs steering committee inputs; signs off phase exit criteria.
- Reviews test approach for architectural alignment; consulted on integration test scope; reviews defect patterns for architectural implications; co-signs Test Plan.
- Clarifies business rules and policy interpretation during test design and execution; consulted on UAT scenarios; supports business defect investigation.
- Engaged on as-needed basis throughout the project (not full-time); consulted on technical queries during SIT and UAT; supports defect investigation for legacy-system integration concerns.
- Execute UAT scripts; validate that the system supports real-world casework; raise UAT defects; sign UAT acceptance.
- Issues DCC Reference Numbers; controls version history of Test Plan and all test deliverables; manages document distribution and retention per MDEC Record Management Policy.

2.1.2. TCS Roles

Key Responsibilities

- Owns TCS delivery commitments; signs off TCS-side phase outputs; primary escalation point for TCS resourcing and delivery issues; co-signs Test Plan from delivery side.
- Co-owns Test Plan review; ensures architectural completeness of integration tests; provides design clarifications during test design; consulted on FAT scope.
- Owns this Test Plan; defines strategy, schedule, and resourcing; chairs defect triage; reports test status; enforces entry/exit criteria; signs off SIT and FAT.
- Designs test scenarios per business module; authors test cases for complex flows (Return/Resubmission, Rejection, TOE, AMP); leads exploratory testing; mentors QA engineers.
- Execute functional test cases across Company Mgmt, Application Mgmt, Payment, Meeting Mgmt, Endorsement & Issuance modules; log and verify defects.
- Builds and maintains regression automation (API + UI); integrates suites into CI; produces automation coverage reports.
- Designs and executes load, stress, and endurance tests; profiles workflow engine and integration layer; reports against SLA targets.

- Ensures unit-test entry criteria are met before SIT handover; commits to defect-fix turnaround; participates in triage; provides build notes and impact analysis.
- Clarifies business rules and policy interpretation; provides Acceptance Criteria for each user story; participates in triage; co-leads UAT with end users.
- Owns Document Migration (OSS → OSS) and Data Migration (MySQL → PostgreSQL) execution and verification per S5.1.1 / S5.1.2; provides migration test outputs and reconciliation reports; co-owns Risk #6 (Migration Sequencing — Dangling File Pointers).
- Execute UAT scripts; validate that the system supports real-world casework; raise UAT defects; sign UAT acceptance.
- Provisions and maintains SIT, UAT, FAT environments; manages deployments, configurations, integration mocks, and test data refresh.
- Owns UAT user training, business readiness, and adoption support during UAT execution and Go-Live preparation; ensures named UAT users are available and trained per S3.0 schedule.

2.1.3. External

Key Responsibilities

- Conducts vulnerability assessment, RBAC verification, and OWASP Top 10–aligned testing during the Penetration Testing window per S3.0; supports remediation revalidation.

2.2. Raci Matrix

The following matrix defines accountability for each major test activity across the FTMS Phase 1 (FKW) test programme. The matrix uses standard RACI conventions:

- **R — Responsible:** Performs the work; the "doer". Multiple people may share R for an activity.
- **A — Accountable:** Owns the outcome; signs off completion. Only one role holds A per activity.
- **C — Consulted:** Provides input before the activity is completed; in the loop and active.
- **I — Informed:** Receives updates on progress and outcomes; not directly involved in the work.
- **— (dash):** No formal role in this activity.

Legends :

TCS	MDEC
TM : Test Manager / QA Lead	PS : Programme Sponsor
SrQA : Senior QA Engineer	PgM :Programme Manager (MDEC)
QAE : QA Engineers (Functional + Automation)	SA : Solution Architect (MDEC)
ML : Migration Lead	BU : Business / UAT Users
Dev : Development Lead	Sec : Security Tester
BA : Business Analyst Lead	DCC : Document Control Centre
OCM : OCM Lead	
DvO : DevOps	
SA : Solution Architect (TCS)	
PgM : TCS Programme Manager	

2.2.1. RACI by Test Activity

#	Test Activity	Ps	PgM	SA	TM	SrQA	QAE	ML	Dev	BA	OCM	BU	DvO	Sec	DCC
1	Master Test Plan authoring	I	C	C	A/R	C	I	C	C	C	I	I	I	C	I
2	SIT Test Plan authoring	I	C	C	A	R	C	C	C	C	I	I	I	I	I
3	UAT Test Plan authoring	I	C	C	A	R	I	I	I	C	C	C	I	I	I
4	Test Case design — Functional	I	I	C	A	R	R	I	C	C	I	I	-	-	I
5	Test Case design — Integration	I	I	C	A	R	R	C	C	C	I	-	C	-	I
6	Test Case design — Migration	I	I	C	A	R	R	R	C	I	I	-	C	-	I
7	Test Case design — UAT	I	I	I	A	R	C	I	I	R	C	C	-	-	I
8	Test data preparation	I	I	C	A	R	R	C	C	C	I	-	C	-	-

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9	Test environment provisioning	I	C	I	C	I	I	C	C	I	I	-	A/R	-	-
10	Unit Testing execution	-	I	C	I	-	-	I	A/R	-	-	-	-	-	-
11	SIT execution — Internal	I	I	C	A	C	R	I	C	C	I	-	C	-	-
12	SIT execution — External Integrations	I	C	C	A	R	R	I	C	C	I	-	C	I	-
13	UAT execution	I	C	I	A	C	C	I	C	C	R	R	-	-	-
14	UAT user training & readiness	I	C	I	C	I	I	I	I	C	A/R	C	I	-	-
15	Penetration Testing	I	C	C	C	I	I	I	C	I	I	C	-	A/R	-
16	Final Acceptance Testing (FAT)	I	C	C	A	R	R	C	C	C	C	C	C	C	-
17	Document Migration verification (OSS → OSS)	I	C	C	A	C	C	R	C	I	I	-	C	-	-
18	Data Migration verification (MySQL → PostgreSQL)	I	C	C	A	C	C	R	C	I	I	-	C	-	-
19	Defect logging	I	I	I	A	R	R	I	C	C	I	R	I	C	-

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20	Defect resolution	I	I	C	C	I	I	C	A/R	C	I	I	C	C	-
21	Defect re-testing & closure	I	I	I	A	R	R	C	C	C	I	C	I	C	-
22	Daily Defect Triage	I	I	I	A/R	R	C	C	R	R	I	C	I	C	-
23	Weekly Status Reporting	C	C	I	A/R	C	I	I	I	I	I	I	I	I	I
24	Phase Closure Report	I	C	C	A/R	R	C	C	C	C	C	C	C	C	I
25	Cutover Readiness Review	A	R	C	R	C	I	C	C	C	C	C	C	C	I
26	Go-Live Decision	A	R	C	C	I	I	C	C	C	C	C	C	I	-
27	Document Control & repository management	I	I	I	C	I	I	I	I	I	I	I	I	I	A/R

2.2.2. Notes on the Matrix

The following clarifications apply to the RACI assignments above:

1. **Single Accountable, multiple Responsible.** Each row has exactly one Accountable role. Where multiple R's appear (e.g. Test Case design — Functional has both Senior QA and QA Engineers as R), this reflects work shared across the team while accountability remains with the Test Manager.
2. **A/R combined notation.** Where a role is shown as **A/R**, the same role is both Accountable and Responsible — typical for activities a single function fully owns (e.g. Unit Testing for Dev Lead, Document Control for DCC).
3. **Project Sponsor accountability.** The Project Sponsor is Accountable only for Cutover Readiness Review and Go-Live Decision. This reflects the executive-sponsor model: oversight and final-call authority on release-impacting decisions, not day-to-day involvement.
4. **External vendors.** Where Penetration Testing or external integration testing involves third-party vendors, those vendors operate under the named MDEC role (Security Tester for Pen Test; Integration Lead for external systems).

2.3. System Personas

The FTMS Phase 1 (FKW) system serves four distinct user personas. Each persona has distinct responsibilities, system access, and test coverage requirements.

#	Persona	Type	Description	Primary Test Coverage
1	Company User	External	Authorised representative of a company subscribed to FTMS. Submits FKW applications on behalf of foreign employees, manages dependants, makes payments, tracks application status, and initiates post-approval transactions.	Company Registration, Activation, FKW Application (Principal & Dependant), Payment 1 / Payment 2, Document Upload, TOE / AMP / Cancellation requests, Application Status Tracking
2	Internal User (MDEC Processing Officer)	Internal	MDEC officer responsible for reviewing, verifying, processing, and deciding FKW applications	Application Review, Internal Processing, Document Verification, Return / Resubmission, Meeting / Decision

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			throughout the workflow. Operates across the Internal Processing, Meeting, Decision, and Endorsement stages.	Stage, Endorsement Officer functions, Reject / Defer workflows
3	JIM User (Immigration Officer)	External Agency	Officer from Jabatan Imigresen Malaysia (JIM), integrated with FTMS for immigration approvals, pass issuance, and immigration record synchronisation.	MyIMMS integration, Pass Issuance workflow, Endorsement processing from immigration side, immigration record updates
4	Administrator	Internal	System administrator with elevated rights — manages users, configurations, master data, audit, and access control across the FTMS platform.	Role-Based Access Control (RBAC), User Management, Master Data Management, Audit Trail review, System Configuration changes, Reference Data maintenance

3.0 TEST SCHEDULE

The detailed test schedule is dependent on the FTMS Project Timeline (currently pending). The activity list below establishes the testing milestones; planned dates will be inserted upon receipt of the Project Timeline document.

No	Testing Activities	Date
1.	Draft SIT Test Plan & UAT Test Plan	TBD
2.	Review SIT Test Plan (TCS) and UAT Test Plan (MDEC)	TBD
3.	Consolidate Master Test Plan (this document, baselined as v1.0)	TBD
4.	Sign-Off Test Plan (MDEC)	TBD
5.	Data Migration Plan signoff	TBD
6.	Data Migration Unit Testing	TBD
7.	Data Migration Dry Run on Staging	TBD
8.	Data Validation Report	TBD
9.	Test Case Design & Review (per module)	TBD
10.	Test Environment Readiness (Staging non-prod, smoke test)	TBD
11.	SIT — FTMS Modules	TBD
12.	SIT — 10 External Integrations	TBD
13.	UAT Kickoff & Environment Validation	TBD

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14.	UAT — User & Company Registration	TBD
15.	UAT — FKW Application Testing (Part 1 & 2)	TBD
16.	UAT — Payment & Endorsement Testing	TBD
17.	UAT — Post-Approval & Integration (TOE/AMP/Cancellation)	TBD
18.	UAT — Final Validation, Regression & Walkthrough	TBD
19.	Bug Fixes (post-UAT cycle 1)	TBD
20.	UAT Retesting (with Data Migration)	TBD
21.	UAT Sign-Off (User Acceptance Report)	TBD
22.	Penetration Testing	TBD
23.	Vulnerability Closure	TBD
24.	Pen Test Revalidation	TBD
25.	Final Acceptance Test (FAT)	TBD
26.	Cutover Readiness Review (CRR)	TBD
27.	Test Summary Report Submission (consolidated)	TBD
28.	Go-Live Preparation	TBD
29.	Go-Live (Phase 1: FKW)	TBD

4.0 TEST RISKS AND CONTINGENCIES

The risks below assume that (a) the High-Level Briefing accurately represents the EXPATS scope and behaviour; (b) Process Flow and Project Timeline documents will be supplied within one week of v0.9; (c) Development unit-test coverage of $\geq 70\%$ on changed code will be achieved before each SIT cycle; (d) at least one usable non-production environment is available throughout the test cycles.

No	Potential Risk	Contingency Plan
1.	Policy / Business Rule Volatility Immigration policy and FKW eligibility rules may change mid-test cycle, invalidating test cases and test data.	Maintain a configuration-driven business-rules layer in test cases (rules separated from scripts). Establish a weekly policy-watch checkpoint with the Business Analyst. Reserve 10% schedule buffer per cycle for rule re-baselining and targeted regression. Tag all rule-dependent test cases for fast impact analysis.

2.	Shared-Backend & Same-Platform Migration FTMS Phase 1 (FKW) shares Workflow, Documents, Integrations, and AI services with DE Rantau and MTEP. Even though DE Rantau and MTEP module builds are out of Phase 1 scope, their legacy data still migrates into the FTMS data store as part of the data migration workstream.	(a) Add explicit shared-service contract tests (API-level) independent of UI scenarios. (b) Coordinate with DE Rantau / MTEP migration leads on data integrity for migrated records. (c) For OSS-to-OSS document migration, include bucket access tests, retention policy verification, ACL preservation, and object integrity checks (MD5/SHA hash comparison: source bucket vs target bucket). (d) For the database engine transition (MySQL → PostgreSQL), treat the schema transformation as a first-class integration test — include row counts, data-type mapping, foreign-key integrity, and pointer integrity to the migrated OSS objects. (e) Flag every defect against shared services, migration paths, or non-FKW migrated data with a "cross-module risk" tag for visibility.
3.	Non-Linear Workflow Coverage Gaps EXPATS is explicitly described as "not a linear workflow." Return/Resubmission loops, Rejection branches, TOE and AMP subprocesses, and split Payment 1 / Payment 2 stages create	Adopt state-transition testing and decision-table techniques as primary methods (not optional). Allocate ≥ 20% of test effort to time-boxed exploratory testing focused on alternate flows. Build an end-to-end state diagram in test design and require coverage attestation against every state transition. Include negative-path and

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	combinatorial paths that scripted testing alone may miss.	rejected-application scenarios as mandatory test cases.
4.	Late Inputs — Process Flow & Project Timeline Detailed Process Flow document and Project Timeline are not yet finalised. Late delivery of either compresses test design time and may delay environment readiness, test data preparation, and schedule baseline.	Continue test plan drafting in parallel using High-Level Briefing as the input baseline (this v0.9 draft). Mark dependent sections [PENDING] with clearly defined re-work scope. On receipt of inputs, conduct a 2-day rapid alignment workshop, baseline as v1.0, and re-validate Section 3.0 Test Schedule and Section 5.1 Functional Test Coverage. Pre-stage generic test data and environment requests now to absorb later schedule pressure.
5.	Migration Sequencing — Dangling File Pointers Document migration (OSS → OSS) and Data migration (MySQL → PostgreSQL) are two separate workstreams. If data migration runs ahead of, or interleaves incorrectly with, document migration, the metadata pointer-rewrite step may target OSS object keys that do not yet exist — producing dangling pointers, broken downloads, and silent data-integrity defects only discovered at UAT or in production.	Treat document migration and data migration as ordered phases with explicit gating: data-migration's pointer-rewrite step must not start until the corresponding document batch is fully landed in OSS and reconciled. Implement an OSS-existence check inside the FTMS Data API before any pointer is committed. Add three-way reconciliation (MySQL count = PostgreSQL count = OSS object count) as a mandatory exit gate. Test these as separate verification activities under S5.1.

6.	Compressed SIT Window for 10 External Integrations The project timeline allocates only 5 working days for SIT (21–27 July 2026): 2 days for FTMS-internal integration plus 3 days to test 10 external integrations (MyIMMS, EInvoice, Finance, Payment Gateway, Active Directory, TalentCorp, LHDN, Email, MD System, SSO). At a flat split this is roughly 0.3 days per integration, which is insufficient for meaningful negative-path and error-handling coverage — particularly for systems outside MDEC's direct control where mock-vs-live behaviour may diverge.	(a) Front-load contract-level integration testing during the build phase — each integration must pass automated contract tests in the development environment before SIT. (b) Treat external SIT not as discovery-mode but as confirmation-mode: defects of integration logic should already have been caught earlier. (c) Pre-stage integration test scripts and credentials by mid-July; any partner not ready by then is escalated as a Sev-1 schedule risk. (d) Reserve a contingency day or extend SIT into the UAT week if any external system is found broken on day 1.
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5.0 TEST COVERAGE

Test coverage is organized around two dimensions: Functional Test (Section 5.1) covering the Business Modules and Supporting Modules of EXPATS, and Non-functional Test (Section 5.2) covering quality attributes. The verification methods aligned to the MDEC template are System Integration Testing (SIT), User Acceptance Test (UAT), Final Acceptance Test (FAT).

5.1. Functional Test

Functional testing of the EXPATS module will follow a four-tier progression aligned to the MDEC template:

- Migrated Data Verification — Validation of data and documents migrated from EXPATS 2.0 (Alibaba) into FTMS (Alibaba Cloud). Treated as two ordered workstreams: Document Migration (5.1.1) and Data Migration (5.1.2).
- Unit Test — Owned by Development; entry criterion to SIT is unit-test coverage and pass-rate thresholds (Section 7.0).
- Integration / Connectivity Test — Verifies contracts with payment gateway, immigration / endorsement systems, identity / SSO, notification, and AI/OCR services.
- System Test (SIT) — Full functional verification of EXPATS modules end-to-end in an integrated environment.

Functional coverage by module :

No	Category	Functionalities to be Tested
1.	Company Management	Company registration, activation/renewal/expiry, company profile maintenance, authorised user management, RBAC for company-side users.
2.	FKW Application & Subprocesses	FKW Application creation (Principal & Dependant), pass-type eligibility (EP/DP/PVP/LTSVP), document upload, draft/submit/withdraw, TOE workflow, AMP workflow, Cancellation workflow, applicant profile capture
3.	Processing & Workflow	State transitions across the EXPATS flow; Return / Resubmission loop; routing to Internal Processing, Meeting, Approved, Rejected; SLA timers; assignment and reassignment to processing officers.
4.	Payment	Payment 1 (pre-processing fee) gating submission; Payment 2 (post-approval fee) gating endorsement; refund / reversal handling; receipt generation; payment-gateway integration; failure and timeout handling.
5.	Endorsement & Issuance	Endorsement task creation post-Payment 2; pass issuance; pass record creation in master data; notifications to applicant and company.

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6.	Document Management (Supporting)	Document upload with format/size validation; versioning; secure storage; retrieval; retention; access control aligned with case ownership.
7.	Master Data Management (Supporting)	Reference data integrity (countries, occupations, sectors, pass categories); pass record lifecycle; data consistency across modules.
8.	Integration Layer (Supporting)	Integration contracts with payment gateway, immigration / endorsement systems, identity / SSO, notification services. Includes positive paths, error handling, retry and idempotency.
9.	AI-Assisted Capabilities	Document Extraction (OCR auto-populate accuracy by field); Document Validation (completeness / consistency precision and recall); Decision Support outputs and human-checkpoint enforcement.
10.	Migration Verification	Document Migration (OSS → OSS) integrity and reconciliation per 5.1.1; Data Migration (MySQL → PostgreSQL) fidelity and pointer-rewrite per 5.1.2.

5.1.1. Document Migration Verification (OSS -> OSS)

Documents currently held in Alibaba OSS (EXPATS 2.0) must be transferred to Alibaba OSS (FTMS) without transformation. The FTMS Doc API behaves as a pass-through; the file that lands in OSS must be byte-identical to the file that left OSS.

Verification scope:

- Byte-level integrity — SHA-256 checksum of every source object equals the checksum of the corresponding target object.
- Object-key mapping — every OSS object key has a deterministic, documented mapping to its OSS object key, and the mapping is reversible for audit purposes.
- Large-file handling — streaming and multipart upload behave correctly for files at and above the documented size limit (e.g., ≥ 50 MB scanned PDFs); no truncation, no memory exhaustion.
- Idempotency on retry — re-running the migration job on a partially-completed batch does not produce duplicates, partial files, or corrupted overwrites.
- Failure handling — transient errors (network, throttling, credential rotation) trigger retry with backoff; permanent errors fail loudly to a defect log, not silently.
- Reconciliation — count of objects in source OSS prefix equals count of objects under target OSS prefix; orphans on either side are zero.

5.1.2. Data Migration Verification (MySQL -> PostgreSQL)

Verification scope:

- Row-count fidelity — for every migrated table, count(MySQL source) equals count(PostgreSQL target) per migration batch.
- Type-mapping fidelity — verified mapping for known type differences: AUTO_INCREMENT → IDENTITY/SERIAL; TINYINT(1) → BOOLEAN; DATETIME → TIMESTAMPTZ (with timezone preserved, no silent UTC drift); MySQL JSON → PostgreSQL JSONB; ENUM handled per documented mapping; case-sensitivity of identifiers preserved per convention.
- Foreign-key & relational integrity — every foreign key in PostgreSQL resolves to a real parent row; no orphaned children; no constraint violations on commit.
- File-pointer rewrite — every metadata column that referenced an OSS URL or bucket+key in MySQL has been rewritten to the corresponding OSS URL/key in PostgreSQL, AND that OSS object actually exists at the rewritten location.
- Three-way reconciliation — count(MySQL rows with file pointer) = count(PostgreSQL rows with file pointer) = count(OSS objects). No dangling pointers, no orphan files.
- Audit traceability — every PostgreSQL row carries migration-audit fields (migrated_at, source_system, source_ref carrying original MySQL primary key, migration_batch_id) and the source_ref resolves back to a real source row.
- Idempotency on retry — re-running a batch does not duplicate rows or revert previously-corrected values.

5.1.3. Unit Test

- Accountability: Explicitly state that this phase is owned and executed by the Development Lead and the development team.
- Validation of individual components, subroutines, and functions in isolation.
 - Verification of internal logic, data processing, and error handling at the code level.
- Entry/Exit Criteria for QA: Code Coverage: Target $\geq 70\%$ coverage on all new or changed code.
 - Unit Test Report: A summary report or checklist confirming 100% pass rate of unit tests is a mandatory entry criterion for SIT.
- Tooling: Mention the use of automated unit testing frameworks (e.g., Selenium).

5.1.4. Integration / Connectivity Test

- Integration Scope: Focused on the 10 external integrations (MyIMMS, Payment Gateway, LHDN, etc.) identified in your plan.
- Verification Points:
 - API Contracts: Ensuring data sent from FTMS is correctly received and acknowledged by external agency endpoints.
 - Connectivity: Verifying stable TLS 1.2+ secure connections and firewall whitelisting between the Staging environment and external sandboxes.
 - Authentication: Successful handshake using Active Directory or SSO credentials.
- Strategic Note: State that these tests are performed in the Staging Environment using mock or stubbed integrations where live sandboxes are unavailable.

5.1.5. System Test

No	Category	Functionalities to be Tested
1.	Company Management	Company registration, activation/renewal/expiry, company profile maintenance, authorised user management, RBAC for company-side users.
2.	FKW Application & Subprocesses	FKW Application creation (Principal & Dependant), pass-type eligibility (EP/DP/PVP/LTSVP), document upload, draft/submit/withdraw, TOE workflow, AMP workflow, Cancellation workflow, applicant profile capture
3.	Processing & Workflow	State transitions across the FKW lifecycle; Return/Resubmission loops; routing to Internal Processing, Meeting/Decision, Approved, Rejected; SLA timers; case assignment and reassignment
4.	Payment	Payment 1 (pre-processing fee) gating submission; Payment 2 (post-approval fee) gating endorsement; refund / reversal handling; receipt generation; payment-gateway integration; failure and timeout handling.
5.	Endorsement & Issuance	Endorsement task creation post-Payment 2; pass issuance; pass record creation in master data; notifications to applicant and company.

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6.	Document Management (Supporting)	Document upload with format/size validation; versioning; secure storage; retrieval; retention; access control aligned with case ownership.
7.	Master Data Management (Supporting)	Reference data integrity (countries, occupations, sectors, pass categories); pass record lifecycle; data consistency across modules.
8.	Integration Layer (Supporting)	Integration contracts with payment gateway, immigration / endorsement systems, identity / SSO, notification services. Includes positive paths, error handling, retry and idempotency.
9.	AI-Assisted Capabilities	Document Extraction (OCR auto-populate accuracy by field); Document Validation (completeness / consistency precision and recall); Decision Support outputs and human-checkpoint enforcement.
10.	Migration Verification	Document Migration (OSS → OSS) integrity and reconciliation per 5.1.1; Data Migration (MySQL → PostgreSQL) fidelity and pointer-rewrite per 5.1.2.
11.	Non-Functional	Vulnerability (OWASP Top 10), Performance (load/stress/endurance/spike), Usability, Reliability & Recovery, Compatibility (browser matrix). See §5.2 for detailed scope.
12.	Detailed Scenarios	<i>[PENDING – process flow] Per-state-transition scenarios, decision-table scenarios per pass type.</i>

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5.2. Non-functional Test

Non-functional verification spans the following quality attributes:

No	Category	Scope
1.	Connectivity Test	Verify connectivity to all dependent services (payment gateway, immigration system, identity provider, notification service, AI/OCR service) across SIT, FAT, and UAT environments.
2.	Vulnerability Test	OWASP Top 10 aligned testing; authentication and session management; RBAC enforcement; PII protection in transit and at rest; input validation and injection resistance; secure file upload.
3.	Performance Test	Load test (target concurrent applications and reviewers), stress test (breaking-point analysis), endurance test (steady load over time), spike test (sudden traffic surge). SLA targets to be confirmed during test design.
4.	Usability	Workflow clarity for company-side users and processing officers; error-message quality; accessibility considerations; mobile/responsive behaviour where applicable.
5.	Reliability & Recovery	Behaviour on integration failure (payment gateway, OCR service, notifications); transactional integrity across multi-step flows; restart / retry / idempotency; backup and restore drills (where in scope).

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6.	Compatibility	Browser compatibility matrix (latest two major versions of Chrome, Edge, Firefox, Safari); document-format compatibility for uploads (PDF, JPG, PNG, etc.).
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5.2.1. Connectivity Test

- **Infrastructure Verification (Infra Test):** Validates the underlying network environment required for FTMS to operate
 - **Network Paths:** Ensuring stable LAN/WAN connectivity between the application layer, databases, and Alibaba OSS storage.
 - **Security:** Verification of firewall whitelisting and secure TLS 1.2+ tunnels for all outbound requests.
 - **Performance Baselines:** Monitoring integration endpoint latency to ensure it remains below 500 ms for synchronous calls.
- **Application Integration (Applicable Test):** Verifies the "contracts" and data exchange with external modules and agencies.
 - **External Systems:** Testing connectivity and data flow for all 10 external integrations, including MyIMMS, Payment Gateway, LHDN, and TalentCorp.
 - **Service Reliability:** Testing the system's behavior during integration failures or timeouts to ensure transactional integrity.
 - **Personas:** Ensuring the Integration Layer correctly handles requests from all four system personas (Company, Internal, JIM, Admin).

6.0 TEST DATA

Test data for the EXPATS module shall be a curated combination of the following sources:

- Synthetic data: Generated covering the full pass-type matrix (EP, DP, PVP, LTSVP) × applicant nationality × dependant relationship × document-quality variants. Used for SIT and FAT.
- Anonymised production-derived data: Where permitted by data-protection policy, sample real-world cases (Personally Identifiable Information masked) used for representative coverage and AI ground-truth datasets.
- Live (production) data: Not used in non-production environments for routine testing. Limited, controlled production-data testing only with documented data-protection approval and only in a hardened pre-production environment.

Test Data Principles:

- Personally Identifiable Information (PII) and passport data shall be masked or synthetic in all non-production environments.
- Test data sets shall be versioned and refreshable on demand to support repeatable test cycles.
- A documented test-data catalogue shall accompany the Test Cases, mapping each test case to the specific test-data record(s) it requires.
- AI Component Validation requires a separate ground-truth dataset of ≥ 500 documents covering pass-type × nationality × scan-quality variants.

7.0 APPROACH

Testing will follow a risk-based, phased approach progressing from Unit Test (Development-owned) through Integration / Connectivity, System Integration Testing (SIT), Final Acceptance Test (FAT), Performance, Security, AI Component Validation, and finally User Acceptance Test (UAT).

7.1. Test Approach

The 9 module categories defined in S5.1.5 (Company Management; FKW Application & Subprocesses; Processing & Workflow; Payment; Endorsement & Issuance; Document Management; Master Data & Administration; Integrations; AI-Assisted Capabilities), plus Migration Verification and Non-Functional scopes as cross-cutting verification workstreams.

The four personas defined in §2.3 (Company User, Internal User, JIM User, Administrator)

Test Design Workflow

1. Input. URS, SRS, Process Flow, and persona definitions baseline the test scope.
2. Define test requirements. Map each requirement to one or more module-persona pairings.
3. Establish test scenarios. Group related requirements into business scenarios for each persona.
4. Establish test cases. Author detailed test cases (preconditions, test steps, test data, expected results) under each scenario.
5. Peer review. Senior QA reviews test cases for completeness, redundancy, and traceability before approval.
6. Storage. Approved test cases are stored in the test management tool with linked URS/SRS references.

Test Execution — Daily Cadence

Test execution operates on a structured daily rhythm during SIT, UAT, and FAT phases:

- Morning standup (15 min) — Test team reviews the day's planned test scenarios, surfaces blockers from prior day, confirms environment readiness and test data state. Held at the start of each test day.
- Test execution — QA Engineers execute assigned test cases. Defects are logged immediately upon discovery and routed to the Development Lead per the Defect Priority and Resolution Fix Time schedule (S8.2.2).

Automation Strategy

Automation is targeted at repetitive scenarios where manual execution provides diminishing returns:

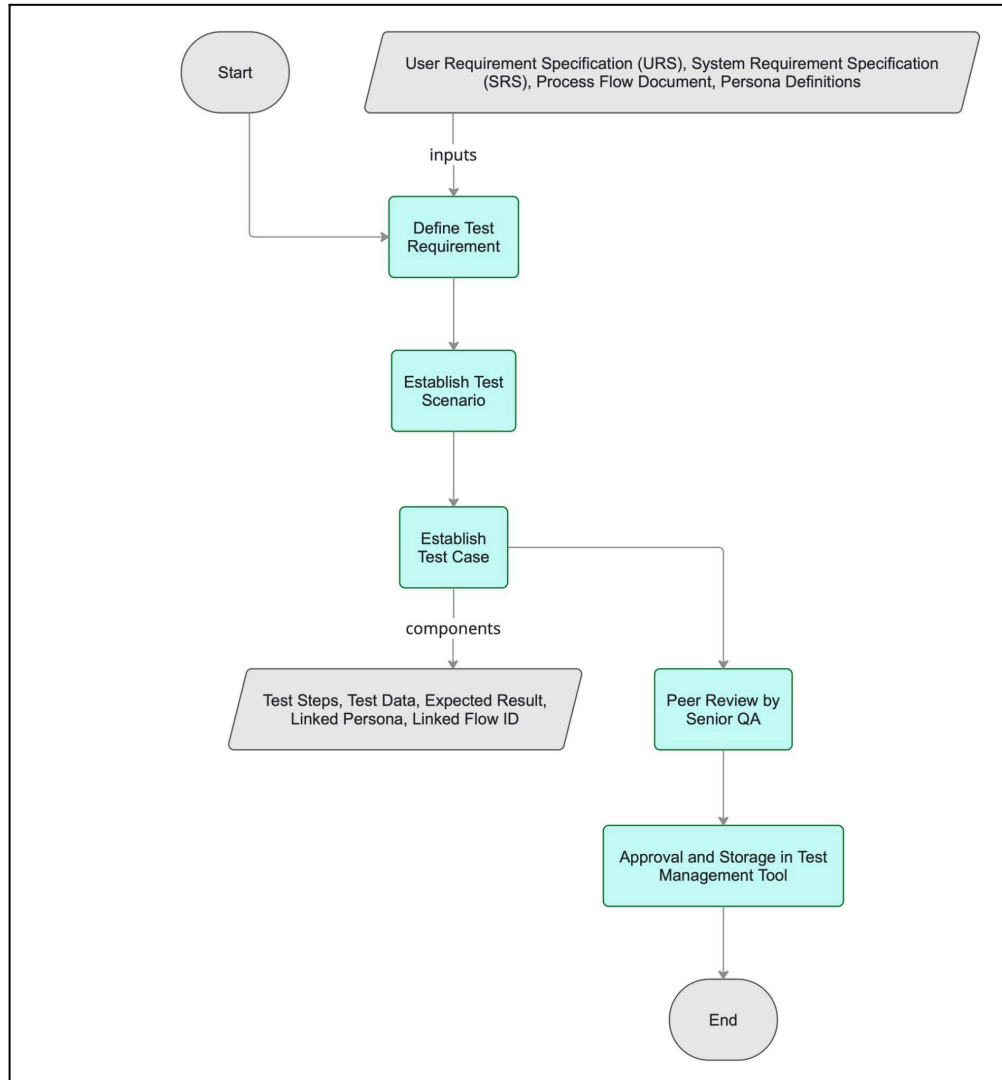
- Functional UI automation: Selenium-based test scripts for repetitive functional flows (login, application form fills, navigation paths).
- API automation: Equivalent API-level test tooling (Postman / Newman or equivalent) for integration tests where UI automation is inappropriate.
- Regression suite: Stable functionality is progressively automated for regression. Target: $\geq 70\%$ automated coverage of Severity-1 and Severity-2 scenarios by UAT start.

Test Process

Three core processes apply across all test levels:

1. Test Planning — establishes scope, schedule, resourcing, and acceptance criteria (this Master Test Plan)
2. Test Development — designs scenarios and test cases from URS / SRS / Process Flow following the workflow above
3. Test Execution — runs test cases, logs defects, manages defect lifecycle per the daily cadence above

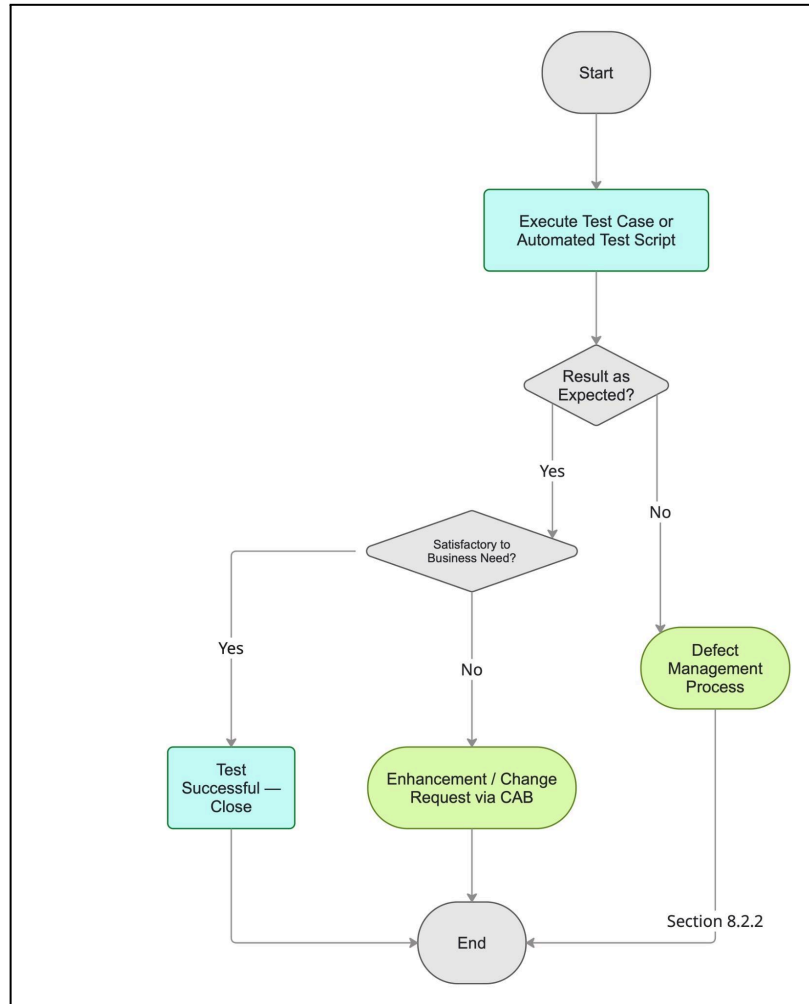
7.2. Test Development Process



Test development for FTMS Phase 1 (FKW) takes the User Requirement Specification (URS), System Requirement Specification (SRS), Process Flow document, and the Persona Definitions (S2.3) as inputs. Each test case must reference its source requirement, the persona executing it, and the relevant Process Flow ID, supporting traceability through to the VCRM (S11.0). Test cases are peer-reviewed by the Senior QA before approval and storage in the test management tool

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7.3. Test Execution Process



For each test case execution: if the result matches the expected outcome AND it satisfies the underlying business need, the test is marked successful and closed. If the result does not match the expected outcome, a defect is raised and routed through the Defect Management workflow (S8.2.2). If the result matches the expected outcome but does not satisfy the underlying business need, a Change Request is raised through the project's Change Advisory Board for assessment as an enhancement, not a defect.

7.4. Test Reporting and Communication

7.4.1. Test Reports

The following reports shall be produced and circulated during active test phases. The Test Manager / QA Lead is accountable for the production and distribution of all test reports.

Report	Description	Audience	Key Metrics
Daily Test Execution Status	Summary of test scenarios executed during the day, broken down by module and persona. Highlights blockers and dependencies.	Test Team, Development Lead, Business Analyst, Project Manager	• Number and percentage of test cases passed • Number and percentage failed • Number and percentage executed (vs planned) • Number blocked (with reason)
Daily Defect Status Summary	List of newly raised, in-progress, and closed defects, broken down by module and severity.	Test Team, Development Lead, Project Manager	• Defects raised today • Defects closed today • Open defects by severity (S1–S5) • Open defects by module • Open defects by status (New / In Progress / Fixed / Re-test / Closed)
Weekly	Consolidated	Test Manager,	• Test execution:

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Status Report	weekly summary of testing activities, progress against plan, key issues, risks, and constraints.	Project Manager, MDEC Project Sponsor, Development Lead, Business Analyst, Lead UAT Business User	cumulative pass / fail / blocked • Defect ageing (defects open longer than Resolution Fix Time targets) • Defects by module and severity • Risks and issues raised this week • Phase milestones tracked vs plan
SIT Closure Report	Final report at the conclusion of each test phase, summarising scope executed, results, defects found, deferred items, and Exit Criteria compliance.	Test Manager, Project Manager, MDEC Project Sponsor, MDEC Business Owner, all stakeholders	• Total scenarios executed • Final pass rate • Final open / deferred defects with severity • Exit Criteria compliance attestation • Lessons learned
Migration Validation Report	Reconciliation report for Document Migration (OSS	Test Manager, Data Migration Lead, Project Manager,	• Source vs target row counts (3-way reconciliation) • Pointer integrity (OSS

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	→ OSS) and Data Migration (MySQL → PostgreSQL).	MDEC	object existence) • Migration duration • Data anomalies / failures
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7.4.2. Test Meetings

The following meetings shall be conducted during the test programme. The Test Manager / QA Lead is responsible for scheduling, facilitating, and capturing minutes for all test meetings.

Meeting	Purpose	Participants	Outputs
Daily Defect Triage	Review newly raised defects, confirm severity and priority assignments, assign defects to fixers, track ageing against Resolution Fix Time, and escalate breaches.	Test Manager, Senior QA, Development Lead, Business Analyst, lead UAT Business User (during UAT)	• Updated defect register • Escalation list • Action items

Weekly Test Status Meeting	Review Weekly Status Report, discuss progress vs plan, agree on the upcoming week's focus, and address risks and issues.	Test Manager, Project Manager, Development Lead, Business Analyst, MDEC representative	• Updated test schedule • Risk and issue updates • Decisions and action items
SIT Kickoff Meeting	Confirm Entry Criteria are met, walk through the phase scope, scenarios, and success criteria, and align stakeholders on the phase plan.	Test Manager, Project Manager, Development Lead, Business Analyst, all assigned testers, MDEC representative	• Phase Entry Criteria sign-off • Confirmed phase plan
Phase Closure Meeting	Review the Phase Closure Report, confirm Exit Criteria compliance, agree on deferred items, and obtain phase sign-off.	Test Manager, Project Manager, MDEC Business Owner, Development Lead	• Phase Closure Report sign-off • Exit criteria attestation • Deferred-defect register

7.4.3. Communication Channels

The following channels shall be used for test-related communication:

Channel	Purpose
Email (with project distribution list)	Formal report distribution, sign-offs, escalations
Project Repository (DCC-managed)	All test artefacts (plans, scenarios, results, reports)
Defect Management Tool (e.g Microsoft List)	Defect logging, tracking, and closure
Test Management Tool (e.g Microsoft Excel)	Test case authoring, execution recording, traceability
Project Collaboration Channel (e.g. Microsoft Teams)	Day-to-day team coordination, ad-hoc queries
Stakeholder Briefings (in-person or virtual)	Status walkthroughs with senior stakeholders, ad-hoc as required

7.4.4. Communication Principles

The following principles govern test communication throughout the programme:

1. **No surprises.** Material risks, blockers, or schedule slippage shall be raised in the next available report or meeting — not held back for "the right moment".
2. **Escalation over assumption.** When in doubt, escalate. The Test Manager is the single point of contact for test-related concerns and shall route them appropriately.
3. **Written record.** Every meeting produces minutes; every decision is recorded; every report is archived in the DCC project repository.
4. **Audience-appropriate detail.** Daily reports are operational and detailed; weekly reports are strategic and summarised; phase closure reports are formal and comprehensive.
5. **Timeliness.** Daily reports are issued each working day; weekly reports are issued by end of the week ; SIT closure reports are issued within 5 working days of phase exit.

7.5. Tools and Techniques

Test management, defect tracking, automation, performance, and security tooling will be confirmed during test design. Where MDEC has standard tooling, those will be adopted in preference to project-specific selections.

7.6. Constraints

Specific schedule constraints, environment-availability constraints, and resource-availability constraints will be confirmed at v1.0 baseline upon receipt of the Project Timeline.

8.0 COMPLIANCE AND ACCEPTANCE CRITERIA**8.1. Compliance Criteria****8.1.1. System Integration Testing (SIT) — Entry & Exit Criteria**

Entry Criteria – SIT

Category	Criteria
Coverage	Unit Testing is 100% completed for all in-scope modules. Component-level integration tests have passed in the development environment.
Documentation	User Requirement Specification (URS) and System Requirement Specification (SRS) have been completed, reviewed, and approved by MDEC. SIT Test Plan, SIT Test Scenarios, and SIT scope are defined and reviewed. SIT Test Cases have been peer-reviewed by the Senior QA.
Test Environment	Staging environment provisioned and configured for SIT phase. Test user accounts have been created across all four personas (Company User, Internal User, JIM User, Administrator). Test data has been prepared, loaded, and validated for SIT execution. All 10 external integrations (MyIMMS, E-Invoice, Finance, Payment Gateway, Active Directory, TalentCorp, LHDN, Email, MD System, SSO) are confirmed reachable from the SIT environment, with credentials and connectivity verified.
Migration Readiness	Document Migration (OSS → OSS) and Data Migration (MySQL → PostgreSQL) unit testing has been complete

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Exit Criteria – SIT

Category	Criteria
Coverage	100% of agreed SIT test cases have been executed. All planned integration paths across the 10 external systems have been exercised at least once. All in-scope state transitions (per Process Flow) have been verified.
Quality	Zero open SIT defects with Severity 1 (System Unusable) or Severity 2 (Major Functional Impact). Open SIT defects with Severity 3 (Moderate) are below 10% of total defects raised. Open SIT defects with Severity 4 (Minor) and Severity 5 (Cosmetic) are below 20% of total defects raised. All Severity 1 and Severity 2 defects identified during SIT have been resolved and re-tested.
Documentation	SIT Test Scenarios, Test Results, Test Execution Log, and SIT Test Closure Report have been reviewed and signed-off by the Test Manager and Project Manager. Any deferred or unresolved defects are formally documented with severity, workaround, and risk assessment, and signed-off by relevant stakeholders.

8.1.2. User Acceptance Testing (UAT) — Entry & Exit Criteria**Entry Criteria – UAT**

Category	Criteria
Coverage	SIT execution is 100% complete and SIT Exit Criteria have been met and signed-off.
Quality	Zero open SIT defects with Severity 1 or Severity 2. SIT defects with Severity 3 are below 10% of total. SIT defects with Severity 4 and Severity 5 are below 20% of total.
Documentation	SIT Test Closure Report has been reviewed and signed-off. UAT Test Plan, UAT Test Scenarios, UAT Test Cases, and UAT Test Data have been prepared, peer-reviewed, and shared with Business Users at least 5 working days prior to UAT execution. UAT Test Scenarios have been reviewed and agreed by Business Users prior to UAT kickoff.
Test Environment	Staging environment refreshed and reconfigured for UAT phase, with all SIT artefacts cleared. Test user accounts created for all named UAT users across the four personas. UAT test data has been loaded and validated. All 10 external integrations are operational from the UAT environment.
Business Readiness	UAT users have been identified, trained on the test process, have allocated their time per the UAT schedule.

Exit Criteria – UAT

Category	Criteria
Coverage	100% of agreed UAT test cases have been executed. All four personas have completed their assigned scenarios. All critical end-to-end business flows (Company Sign-up → FKW Application → Internal Processing → Decision → Endorsement → Pass Issuance, plus TOE/AMP/Cancellation subprocesses) have been validated.
Quality	Zero open UAT defects with Severity 1 or Severity 2. Open UAT defects with Severity 3 are below 10% of total UAT defects. Open UAT defects with Severity 4 and Severity 5 are below 20% of total UAT defects. All Severity 1 and Severity 2 defects identified during UAT have been resolved, re-tested, and signed-off by the originating Business User.
Documentation	UAT Test Scenarios, Test Results, UAT Defect Log, and UAT Test Closure Report (User Acceptance Report) have been reviewed and signed-off by the Business Users, Test Manager, and Project Manager. Any deferred or unresolved defects are formally documented with severity, workaround, business-impact assessment, and signed-off by MDEC as acceptable for production release.
Sign-off	Formal UAT Sign-Off has been obtained from MDEC Business Owner.

8.1.3. Final Acceptance Testing (FAT) — Entry & Exit Criteria

Entry Criteria – FAT

Category	Criteria
Coverage	UAT execution is 100% complete, UAT Exit Criteria have been met, and UAT Sign-Off has been obtained.
Quality	All defects identified during UAT have been resolved or formally deferred with documented justification. Penetration Testing has commenced (FAT runs in parallel with Pen Test revalidation per project timeline).
Documentation	UAT Test Closure Report and UAT Sign-Off are available. FAT scope has been agreed between TCS and MDEC. The FAT scenario list (focused on critical end-to-end business flows and Production-readiness validation) has been finalised.
Test Environment	The Pre-Production / Staging environment is fully configured to Production specification. Production-equivalent data set has been loaded. All integrations are operational against Production-configured endpoints (where applicable).

Exit Criteria – FAT

Category	Criteria
Coverage	100% of agreed FAT scenarios executed and validated. All Production-readiness checks (cutover dry-run, rollback procedure, data migration final validation) have been performed and confirmed.
Quality	Zero open FAT defects of any severity. Any defect found during FAT must be resolved before sign-off, or formally deferred with explicit MDEC approval and a documented workaround.
Documentation	FAT Test Report has been produced, reviewed, and signed-off by the Test Manager, Project Manager, and MDEC representative. Any deferred items are recorded in the Cutover Readiness Review (CRR) input pack.
Sign-off	Formal Final Acceptance Sign-Off has been obtained, authorising progression to the Cutover Readiness Review and Go-Live preparation.

8.1.4. Cross-Phase Compliance Notes

The following compliance principles apply across all test phases:

1. **Defect aging.** Any Severity 1 or Severity 2 defect open for more than 5 working days without resolution shall be escalated to the Test Manager and Project Manager.
2. **Re-test policy.** Every defect resolution must be re-tested by the original tester (or designated peer) before being closed. Defect closure is the responsibility of the original tester, not the developer.
3. **Regression scope.** Any defect fix or change introduced after a phase's exit shall trigger a targeted regression test on the affected functional area before that phase can be re-closed.
4. **Documentation standard.** All test artefacts (scenarios, cases, results, reports, defect logs) shall be stored in the project repository per the Document Control Centre (DCC) standard. Versioned and accessible to MDEC throughout the engagement.

8.2. Acceptance Criteria

8.2.1. Defect Severity

Defects shall be classified using the following severity scale (per the MDEC template):

Severity Level	Description
1	System Unusable
2	Function Unusable
3	Serious
4	Minor
5	Cosmetic Error

8.2.2. Defect Priority and Resolution Time

Priority	Definition	Resolution Fix Time
1	The defect causes complete blockage of a test scenario, an entire functional area, or end-to-end business flow. No further testing of the affected area can proceed until the defect is resolved.	4 working hours
2	The defect has a significant impact on test execution. Some test cases are blocked or producing invalid results, but unrelated areas can continue to be tested.	8 working hours (1 working day)
3	The defect impacts test progress but does not halt it. Affected tests can be deferred or worked around without blocking the overall phase.	16 working hours (2 working days)
4	The defect has only a minor impact on test progress. Typically cosmetic or superficial issues (e.g. mis-spelt labels, mis-aligned UI elements) that do not affect functionality.	40 working hours (1 working week)

9.0 TEST ENVIRONMENTAL NEEDS**9.1. Environment**

The following test environments are required. Each is non-production and isolated from production data:

Environment	Purpose	Test Phases Hosted	Difference from Production
DEV	Developer testing, unit testing	Unit Testing	Lower compute spec; mock integrations; developer-team access only
Staging	System Integration Testing and User Acceptance Testing	SIT (Internal & External Integrations); UAT	Production-like configuration; sandbox / stubbed external integrations; synthetic and anonymised migrated data; QA and UAT participant access
Pre-Prod	Final Acceptance Testing, Penetration Testing, Cutover Readiness	Penetration Testing; FAT; CRR dry-run	Production-equivalent compute and topology; live sandbox integrations (e.g. Immigration sandbox, Payment Gateway sandbox); fully migrated and reconciled data; restricted access to FAT participants and Pen Test

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			vendor
Production	Live production system	Out of scope for test execution	Live system; production data; full RBAC and audit; not used for test cycles

Security level: All test environments require role-based access control (RBAC), audit logging of access, and encryption at rest and in transit. PII data must be masked in all non-production environments.

10.0 TEST DELIVERABLES

No	Deliverable	Description
1.	Test Plan (this document, complete with VCRM)	This document at v1.0 baseline, signed off by Author, Reviewer, and HOD Approver; Appendix A VCRM populated to traceability completeness.
2.	Test Case Script	Detailed test cases for each functionality in Section 5, with steps, test data references, expected results, and traceability to requirement IDs.
3.	Test Observation Report	Per-cycle log of test execution, including pass/fail status, defects raised, and observations against expected behaviour.
4.	Test Summary Report and Test Log	End-of-cycle summary covering coverage, defect statistics by severity, exit-criteria attainment, and Go/No-Go recommendation.
5.	User Acceptance Test Checklist	Business-scenario-based checklist used by UAT participants, organised by EXPATS persona and pass type.

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6.	User Acceptance Test Acknowledgement	Formal sign-off from UAT participants and the Project Sponsor confirming acceptance of the EXPATS module for production release.
7.	Defect Log	Live record of all defects discovered during testing, with severity, status, owner, and resolution evidence; exported as final artefact at project close.
8.	AI Component Validation Report	Quantitative report on Document Extraction (OCR) accuracy, Document Validation precision/recall, and Decision Support performance against acceptance thresholds.
9.	Performance Test Report	Load, stress, endurance, and spike test results with response-time percentiles, throughput, error rates, and SLA attainment.
10.	Security & Vulnerability Assessment Report	OWASP-aligned vulnerability findings, RBAC test results, and PII protection verification.

11.0 APPENDIX A – VERIFICATION CROSS REFERENCE MATRIX (VCRM)

<Foreign Talent Management System> VERIFICATION CROSS REFERENCE MATRIX FORM (VCRM)							
USER REQUIREMENTS SPECIFICATIONS	SRS	SYSTEM FUNCTION	TEST CASE NUMBER	VERIFICATION METHOD	TEST SCOPE		
					UAT	FAT	SIT
[Pending URS]	[Pending SRS]	Company Management: Registration and activation.	[TBD]			(End-to-end)	
[Pending URS]	[Pending SRS]	Application Management: FKW profile capture and pass-type selection.	[TBD]				
[Pending URS]	[Pending SRS]	AI Capabilities: OCR auto-populate and document validation.	[TBD]				

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<i>[Pending URS]</i>	<i>[Pending SRS]</i>	Migration: Document transfer integrity (OSS to OSS).	<i>[TBD]</i>				
Prepared By : Name : Signature : Date :			Verified By : Name : Signature : Date :				